Chapter 15 — Observability and Production

This chapter is about keeping the application working once users depend on it: knowing whether it is healthy, how fast it is, which request failed and why, shutting down without dropping requests, and starting quickly in **containers** (isolated packages of an app plus everything it needs, run by Docker or Kubernetes). Boot's "production-ready features": **Actuator** (management endpoints), **Micrometer** (metrics and the Observation API), **Micrometer Tracing + OpenTelemetry** (distributed tracing: following one request across services), **structured logging**, graceful shutdown (finishing in-flight requests before stopping), virtual threads, and faster startup with AOT (ahead-of-time) caches and native images (§15.11).

The code lives in two Spring Boot 4.1.1 / Framework 7.0 projects on Java 25 (outputs are real `mvn -q verify` output, log noise trimmed as `...`):

- `code/observability-demo/` — Actuator, health, metrics, Prometheus, tracing, structured logs (8 tests).
- `code/production-runtime/` — graceful shutdown and virtual threads (4 tests), plus `code/aot-cache-demo.sh` for the startup comparison.

> **Prerequisites.** Configuration properties and profiles: Chapter 3. Security filter chains: Chapter 10. `RestClient`: Chapter 6. `@ConcurrencyLimit`: 4.7. System-design view of health checks and tracing: Operating in Production.

15.1 Observability in one page

What it is

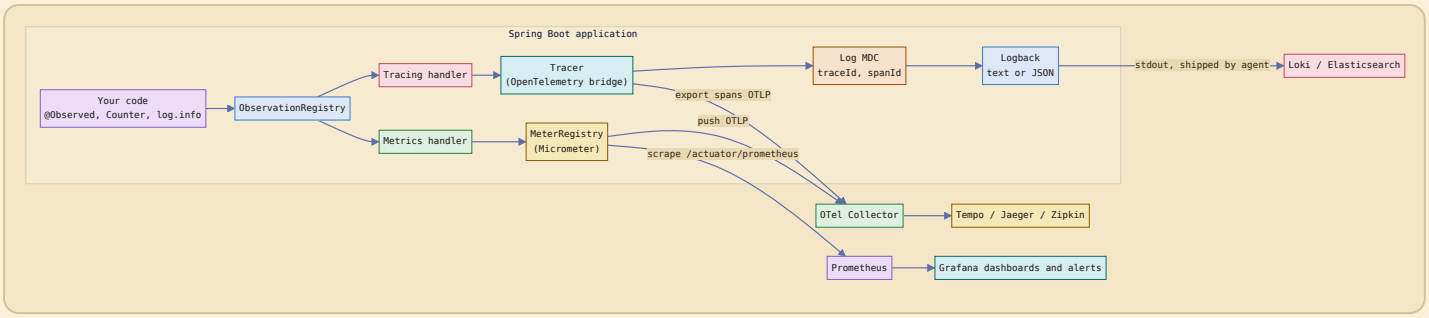
Observability is how well you can understand what a running system is doing from the data it emits — without attaching a debugger or adding code. The data comes in three kinds, the **three pillars**:

Pillar	What it is	Answers	Spring tool
Logs	timestamped text or JSON events	"what exactly happened in this request?"	SLF4J + Logback (Java's standard logging API and Boot's default implementation), structured logging
Metrics	numbers aggregated over time (counts, durations, gauges)	"how many, how fast, how full — right now and as a trend?"	Micrometer `MeterRegistry`
Traces	the path of one request through many services, as a tree of timed **spans** (one span = one timed step of work, §15.7)	"where did this slow request spend its time?"	Micrometer Tracing + OpenTelemetry

Like a hospital patient: **metrics** are the monitor (heart rate over time), **logs** the nurse's notes, the **trace** the record of departments visited and for how long. The monitor says *something* is wrong, the visit record *where*, the notes *what*.

Monitoring is the older, narrower idea: watching known metrics and alerting on thresholds (e.g. "error rate above 1%"). Observability adds asking new questions ("which customer's requests are slow?") from rich, correlated data.

How Spring wires it together



- **Micrometer** is a vendor-neutral metrics **facade** ("SLF4J for metrics"): you write to a `MeterRegistry`; a registry implementation (Prometheus — the popular open-source metrics database, OTLP, Datadog, ...) formats and ships the numbers.
- **Micrometer Observation** is one API producing *both* a metric and a span for a piece of work.
- **Micrometer Tracing** is a facade over a tracer; Boot 4 supports **OpenTelemetry** (`spring-boot-starter-opentelemetry`) and **Brave** with Zipkin (`spring-boot-starter-zipkin`).
- The tracer puts the current `traceId` and `spanId` into the logging **MDC** (Mapped Diagnostic Context, a per-thread map log patterns can print), linking every log line to its trace.

Real-world examples

- Grafana's "LGTM" stack (Loki logs, Grafana, Tempo traces, Mimir/Prometheus metrics) is a common open-source destination for these three outputs.
- The OpenTelemetry Collector is widely deployed as a **sidecar** (helper container next to the app in the same pod) or agent receiving OTLP (the OpenTelemetry protocol) and forwarding to any **backend** (storage + UI), so apps don't hard-code the vendor.

15.2 Spring Boot Actuator

What it is

Actuator adds **management endpoints**: small built-in HTTP (or JMX — Java Management Extensions, the JVM's older management protocol) APIs for operators — health, metrics, info, loggers, environment, thread dumps and more. Add `spring-boot-starter-actuator` and they appear under `/actuator` — the car's service port that mechanics (Kubernetes, Prometheus) plug into and drivers never see.

The endpoints

ID	Shows / does	Notes
`health`	UP/DOWN status, components, probes	the only endpoint exposed over HTTP by default
`info`	build, git, Java, custom `info.*` properties	contributors switched on by properties
`metrics`	list of meters; `/metrics/{name}` for one	JSON, for humans
`prometheus`	all meters in Prometheus text format	needs `micrometer-registry-prometheus`
`loggers`	view and **change** log levels at runtime	`POST /actuator/loggers/com.acme {"configuredLevel":"DEBUG"}`
`env`, `configprops`	resolved properties, `@ConfigurationProperties` beans	values masked by default (`show-values=never`)
`beans`, `conditions`, `mappings`	beans, auto-configuration report, request mappings	great for debugging (Chapter 3)
`threaddump`, `heapdump`	JVM thread dump (what every thread is doing), heap dump file (every object in memory)	heap dumps can contain secrets — protect them
`httpexchanges`	recent requests/responses	needs an `HttpExchangeRepository` bean
`scheduledtasks`, `caches`, `flyway`, `liquibase`, `quartz`, `sessions`	feature-specific	appear when the feature is present
`startup`	startup steps and their duration	needs `BufferingApplicationStartup`
`shutdown`	shuts the app down	access `none` by default

Exposure vs access

Two separate switches decide whether an endpoint can be called:

1. **Access** (since Boot 3.4, replacing the older `enabled` flags): is the endpoint allowed at all, and may it change things? Values `none`, `read-only`, `unrestricted`.
2. **Exposure**: is it reachable over **HTTP** or **JMX**? By default only `health` is exposed on both.

```
# expose more endpoints over HTTP ("*" = all; quote it in YAML)
management.endpoints.web.exposure.include=health,info,metrics,prometheus,env,loggers
management.endpoints.web.exposure.exclude=heapdump

# access control
management.endpoint.loggers.access=read-only        # can read levels, cannot change them
management.endpoints.access.default=none            # opt-in style: everything off ...
management.endpoint.health.access=read-only         # ... except what you list
management.endpoints.access.max-permitted=read-only # hard ceiling for the whole app

# a separate port (and optionally address) for all management endpoints
management.server.port=8081
management.endpoints.web.base-path=/manage          # default /actuator
```

A separate **management port** is common in Kubernetes: the public **ingress** (entry point for outside traffic) routes only to 8080, while probes and Prometheus reach 8081 inside the cluster — keeping operational endpoints off the public path.

Securing actuator endpoints

With Spring Security on the classpath you decide who calls what with `EndpointRequest` matchers. In Boot 4 the class moved to `org.springframework.boot.security.autoconfigure.actuate.web.servlet.EndpointRequest` (Boot 3: `org.springframework.boot.actuate.autoconfigure.security.servlet`).

```
@Bean
SecurityFilterChain security(HttpSecurity http) throws Exception {
    http.authorizeHttpRequests(auth -> auth
            // probes, info and the Prometheus scrape stay open (normally reachable only inside the cluster)
            .requestMatchers(EndpointRequest.to("health", "info", "prometheus")).permitAll()
            // every other actuator endpoint needs the OPS role
            .requestMatchers(EndpointRequest.toAnyEndpoint()).hasRole("OPS")
            .anyRequest().permitAll())
        .httpBasic(Customizer.withDefaults())
        .csrf(csrf -> csrf.disable());   // demo API only; see chapter 10 for CSRF
    return http.build();
}
```

`EndpointRequest` matches by endpoint **ID**, so it survives base-path or port changes. `ActuatorEndpointsTests` proves both sides:

```
anonymous /actuator/env -> 401, ops /actuator/loggers/ch15.obs -> {"effectiveLevel":"INFO"}
```

Common mistakes

- `management.endpoints.web.exposure.include=*` on a public port without security: `env`, `heapdump` and `threaddump` leak secrets and internals. Expose the minimum; protect the rest.
- Thinking exposure is enough — an endpoint with access `none` stays unavailable even when exposed.
- Forgetting that `health` details (`show-details`) can reveal hostnames, disk paths and versions (a disk path appears below).

15.3 Health, health groups and Kubernetes probes

What it is

The `health` endpoint aggregates **health indicators** — small checks like "can I reach the database?" or "is there disk space?" — into one status: `UP`, `DOWN`, `OUT_OF_SERVICE` or `UNKNOWN`, overall the "worst" one (default order `DOWN`, `OUT_OF_SERVICE`, `UP`, `UNKNOWN`). `DOWN` and `OUT_OF_SERVICE` return HTTP **503**, others 200.

Auto-configured indicators appear with their technology: `ping`, `diskSpace`, `db` (a `DataSource`), `redis`, `mongo`, `rabbit`, `mail`, `ssl` (certificate validity), `livenessState`, `readinessState`, and more.

Why it matters

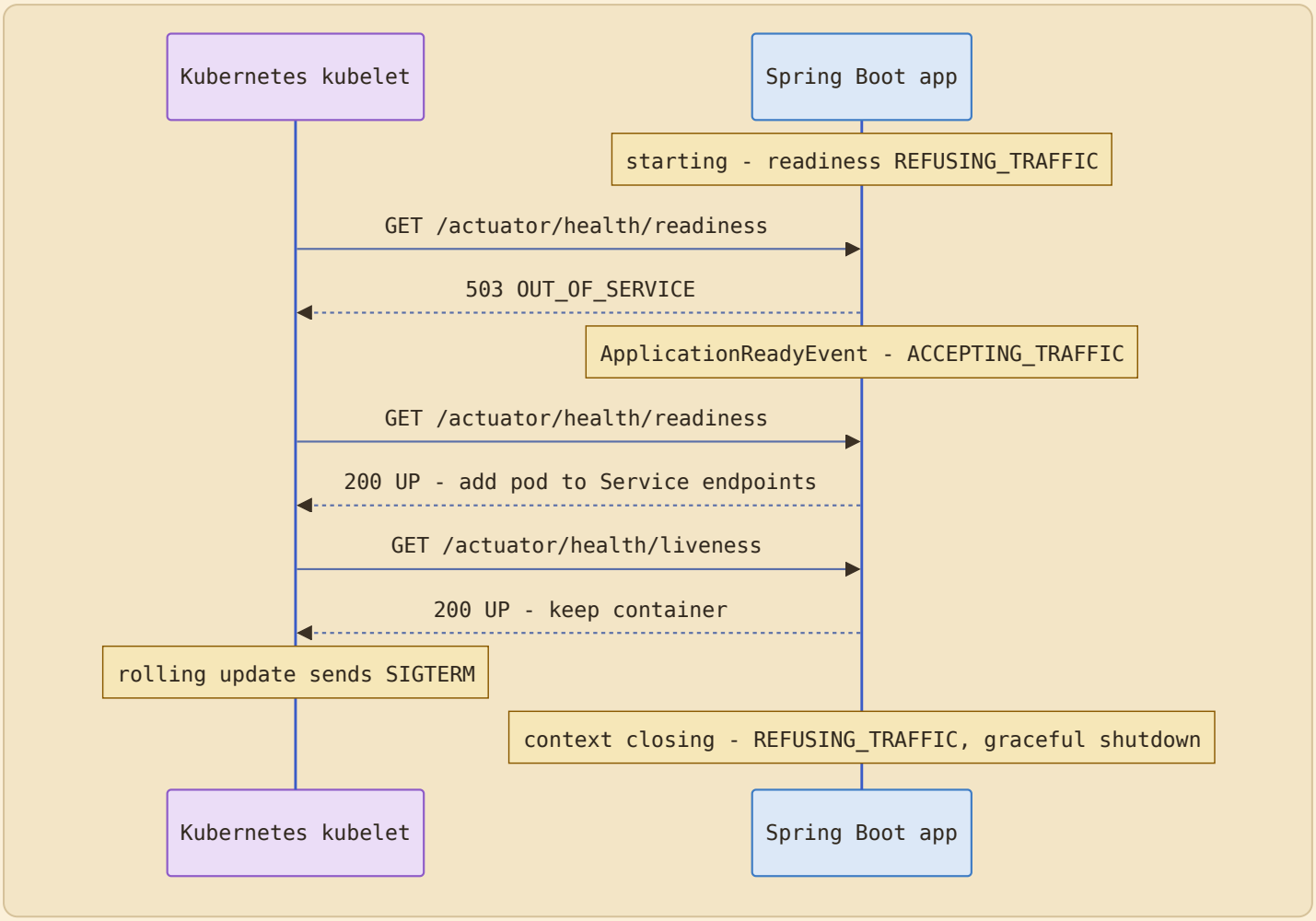
Load balancers and **orchestrators** (Kubernetes) act on these answers: take an instance out of rotation, restart a container, stop a **rolling deployment** (replacing instances a few at a time) — see System Design Fundamentals. A wrong design can **cause** outages: if every instance reports DOWN because one optional dependency is slow, the platform removes all of them.

Liveness vs readiness

Kubernetes asks two different questions:

Probe	Question	On failure Kubernetes...	Should include
Liveness	"Is this process broken beyond repair?"	**restarts** the container	only internal state (deadlock, corrupted state) — never external systems
Readiness	"Can this instance serve traffic right now?"	**stops routing** traffic to it (no restart)	warming up, shutting down, and — carefully — dependencies it cannot work without
Startup (K8s only)	"Has it finished starting?"	holds off the other probes	usually point it at the liveness endpoint with a larger threshold

Boot tracks two **availability states**: `LivenessState` (`CORRECT` / `BROKEN`) and `ReadinessState` (`ACCEPTING_TRAFFIC` / `REFUSING_TRAFFIC`). It marks the app ready only after startup (including `CommandLineRunner`s) finished, and switches readiness to `REFUSING_TRAFFIC` when the context starts closing. **In Boot 4 the probe groups are on by default** — `/actuator/health/liveness` and `/actuator/health/readiness` exist without configuration (Boot 2.3–3.x: only on Kubernetes, or with `management.endpoint.health.probes.enabled=true`).



A custom `HealthIndicator`

In Boot 4 the health API has its own module: `org.springframework.boot.health.contributor.HealthIndicator` and `Health` (Boot 3: `org.springframework.boot.actuate.health`). The component name is the bean name minus the `HealthIndicator` suffix.

```java
// Bean name "inventoryHealthIndicator" -> health component name "inventory"
@Component
public class InventoryHealthIndicator implements HealthIndicator {

    private final InventoryStatus status;   // pretend connection state, flipped by the tests

    public InventoryHealthIndicator(InventoryStatus status) {
        this.status = status;
    }

    @Override
    public Health health() {
        if (status.reachable()) {
            return Health.up().withDetail("latencyMs", 12).build();
        }
        return Health.down().withDetail("error", "inventory service not reachable").build();
    }
}
```

Keep `health()` **fast and cheap** — it runs on every probe (typically every 5–10 s per instance); cache slow checks or compute them in the background. Reactive apps implement `ReactiveHealthIndicator`.

Health groups

A **group** is a named subset of indicators with its own settings, at `/actuator/health/{group}`; liveness and readiness are two built-in groups.

```properties
management.endpoint.health.show-details=when-authorized     # never | when-authorized | always
management.endpoint.health.roles=OPS
# a custom group, e.g. for a deployment smoke check
management.endpoint.health.group.deploy.include=ping,inventory
management.endpoint.health.group.deploy.show-details=always
# readiness also fails when inventory is down (a deliberate choice - see Trade-offs)
management.endpoint.health.group.readiness.include=readinessState,inventory
```

Other group options: `exclude`, `status.order`, `status.http-mapping.*`, `additional-path` (serve the group on the main port, e.g. `server:/healthz`). `management.endpoint.health.probes.add-additional-paths=true` also serves `/livez` and `/readyz` on the main port.

Proof

The tests call the endpoints anonymously and as the `ops` user, then make "inventory" unreachable:

```
anonymous /actuator/health -> {"groups":["deploy","liveness","readiness"],"status":"UP"}
ops       /actuator/health -> {"components":{"diskSpace":{"details":
{"total":494384795648,"free":63519252480,"threshold":10485760,"path":".../observability-
demo/.","exists":true},"status":"UP"},"inventory":{"details":{"latencyMs":12},"status":"UP"},"livenessState":
{"status":"UP"},"ping":{"status":"UP"},"readinessState":{"status":"UP"},"ssl":{"details":{"expiringChains":
[],"invalidChains":[],"validChains":[]},"status":"UP"}},"groups":["deploy","liveness","readiness"],"status":"UP"}
inventory reachable:
  liveness  -> {"status":"UP"}
  readiness -> {"status":"UP"}
inventory DOWN:
  liveness  -> {"status":"UP"}
  readiness -> {"status":"DOWN"} (HTTP 503)
  deploy    -> {"components":{"inventory":{"details":{"error":"inventory service not
reachable"},"status":"DOWN"},"ping":{"status":"UP"}},"status":"DOWN"} (HTTP 503)
```

- Anonymous callers get only status and group names — details need `OPS` (`when-authorized`).
- The `liveness` group exists unconfigured: probes are on by default in Boot 4.
- Liveness stays UP while a dependency is down — restarting our container wouldn't fix the inventory service.

The app can change its own state, e.g. to drain traffic before maintenance:

```java
AvailabilityChangeEvent.publish(context, ReadinessState.REFUSING_TRAFFIC);
```

```
after REFUSING_TRAFFIC: readiness -> {"status":"OUT_OF_SERVICE"} (HTTP 503)
```

Kubernetes configuration (not run here):

```yaml
livenessProbe:
  httpGet: { path: /actuator/health/liveness, port: 8081 }
  periodSeconds: 10
  failureThreshold: 3
readinessProbe:
  httpGet: { path: /actuator/health/readiness, port: 8081 }
  periodSeconds: 5
  failureThreshold: 3
startupProbe:
  httpGet: { path: /actuator/health/liveness, port: 8081 }
  periodSeconds: 5
  failureThreshold: 30        # up to 150 s to start before liveness takes over
```

Trade-offs

A dependency in **readiness** protects users from an instance that can't serve — but if *all* instances share it, all become unready at once and the service vanishes, even for requests not needing it. Include a dependency only if the instance is truly useless without it; otherwise stay ready and **degrade** (fallbacks, cached data, circuit breakers — Chapter 16).

Common mistakes

- Database check in **liveness** → a DB blip restarts every pod, slowing recovery (a "restart storm") — a restart doesn't fix the database.
- Slow indicators (a synchronous remote call per probe) → probe timeouts → false failures.
- Same thresholds for liveness and readiness → a slow start gets killed; use a startup probe.
- `show-details=always` on a public port → leaks infrastructure details.

In the interview

Q: What's the difference between liveness and readiness, and how does Spring Boot support them? 30-second answer: Liveness asks "should the container be restarted?", readiness "should it receive traffic?". Boot tracks `LivenessState` and `ReadinessState`, exposes them as health groups at `/actuator/health/liveness` and `/readiness` (on by default in Boot 4), marks readiness only after startup completes and refuses traffic during shutdown. External dependencies never belong in liveness.

- *Custom check?* → A `HealthIndicator` bean returning `Health.up()/down().withDetail(...)`, included in a group if needed.
- *Take one instance out of rotation without killing it?* → Publish `AvailabilityChangeEvent` with `ReadinessState.REFUSING_TRAFFIC`.

TL;DR

`HealthIndicator` beans → `/actuator/health`; groups (`management.endpoint.health.group.<name>.include`) → `/actuator/health/<name>`; liveness/readiness on by default in Boot 4; DOWN → 503; liveness internal-only, care with dependencies in readiness.

15.4 The info endpoint

`/actuator/info` shows facts about the running build — "which version is actually deployed?" — assembled from **info contributors**:

Contributor	Content	Enabled by
`build`	group, artifact, version, build time	present when `META-INF/build-info.properties` exists (Maven plugin goal `build-info`)
`git`	branch, commit id and time	present when `git.properties` exists (`git-commit-id-maven-plugin`)
`env`	every `info.*` property	`management.info.env.enabled=true`
`java`	JVM and runtime vendor/version	`management.info.java.enabled=true`
`os`, `process`	OS name/version, PID (process id), CPUs, memory	`management.info.os.enabled`, `management.info.process.enabled`
custom	anything	an `InfoContributor` bean

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <executions>
    <execution><goals><goal>build-info</goal></goals></execution>
  </executions>
</plugin>
```

```
management.info.env.enabled=true
management.info.java.enabled=true
info.app.description=Orders demo for chapter 15
```

```
/actuator/info -> {"app":{"description":"Orders demo for chapter 15"},"build":{"artifact":"ch15-observability-demo","name":"ch15-observability-demo","time":"2026-09-28T23:50:25.503Z","version":"1.0.0","group":"handbook.spring"},"java":{"jvm":{"name":"OpenJDK 64-Bit Server VM","vendor":"Homebrew","version":"25.0.2"},"runtime":{"name":"OpenJDK Runtime Environment","version":"25.0.2"},"vendor":{"name":"Homebrew","vers...
```

Never put secrets in `info.*` — the endpoint is usually public.

15.5 Metrics with Micrometer

What it is

A **metric** is a number measured repeatedly over time. Micrometer records them in **meters** in a `MeterRegistry`. Each meter has a **name** and **tags** (key/value dimensions, e.g. `status=200`); every unique name + tag-values combination is its own **time series**.

Meter	Measures	Example	Prometheus output
`Counter`	a count that only increases	orders placed, errors	`name_total`
`Timer`	count + total time + max of short events	request latency	`name_seconds_count`, `_sum`, `_max` (+ buckets if histogram on: counts per value range, e.g. "under 100 ms")
`Gauge`	a current value sampled when read	queue size, pool usage	`name`
`DistributionSummary`	distribution of non-time values	payload size	`name_count`, `_sum`, `_max`
`LongTaskTimer`	tasks *currently* running and for how long	a batch job in progress	`name_active_count`, `_duration_sum`

Like a car: odometer = **counter** (only increases; rate over time), fuel needle = **gauge** (current value, up and down), lap stopwatch = **timer**.

What Boot gives you for free

With Actuator on the classpath, Boot registers JVM memory/GC/threads/classes, CPU, uptime, Tomcat sessions, HikariCP pool, logback events, `http.server.requests` (every MVC/WebFlux request, tagged `method`, `uri` template, `status`, `outcome`, `exception`), `http.client.requests` (for `RestClient`/`WebClient` built from Boot's builders), cache, Kafka and more.

Custom meters

From `OrderService.java`:

```
public OrderService(MeterRegistry registry) {
    this.registry = registry;
    // A Timer records how many calls and how long they took
    this.placeTimer = Timer.builder("orders.place")
            .description("time to place an order")
            .register(registry);
    // A Gauge samples a current value on demand (here: the map size)
    registry.gauge("orders.stored", orders, Map::size);
}

@Observed(name = "order.lookup", contextualName = "order-lookup")   // see 15.6
public Optional<String> find(long id) {
    log.info("looking up order {}", id);
    Optional<String> item = Optional.ofNullable(orders.get(id));
    // A Counter only goes up; the tag splits it into one time series per result
    Counter.builder("orders.lookups")
            .tag("result", item.isPresent() ? "found" : "missing")
            .register(registry)            // returns the existing counter if already registered
            .increment();
    return item;
}

public long place(String item) {
    return placeTimer.record(() -> { /* ... store it ... */ });
}
```

Common tags for every meter come from properties:

```
management.metrics.tags.region=eu-west-1
# optional: publish histogram buckets so Prometheus can compute p95/p99 across instances
management.metrics.distribution.percentiles-histogram.http.server.requests=true
```

Reading metrics: `/actuator/metrics` and `/actuator/prometheus`

The test makes two successful lookups, one miss and one order, then reads both endpoints:

```
/actuator/metrics/orders.lookups -> {"availableTags":[{"tag":"result","values":["missing","found"]},
{"tag":"region","values":["eu-west-1"]}],"measurements":
[{"statistic":"COUNT","value":3.0}],"name":"orders.lookups"}
/actuator/prometheus (selected lines):
  http_server_requests_seconds_count{error="none",exception="none",method="GET",outcome="CLIENT_ERROR",region="eu-
west-1",status="404",uri="/orders/{id}"} 1
  http_server_requests_seconds_count{error="none",exception="none",method="GET",outcome="SUCCESS",region="eu-west-
1",status="200",uri="/orders/{id}"} 2
  order_lookup_seconds_count{class="ch15.obs.OrderService",error="none",method="find",region="eu-west-1"} 3
  orders_lookups_total{region="eu-west-1",result="found"} 2.0
  orders_lookups_total{region="eu-west-1",result="missing"} 1.0
  orders_place_seconds_count{region="eu-west-1"} 1
  orders_stored{region="eu-west-1"} 3.0
```

- `/actuator/metrics/{name}` aggregates across tags (COUNT 3.0); drill down with `?tag=result:found`.
- **Naming** is translated per backend: Micrometer's dotted `orders.lookups` becomes Prometheus `orders_lookups_total` (counter suffix); timers become `_seconds_*`.
- The `uri` tag is the **template** `/orders/{id}`, not `/orders/1` — otherwise every id would be a new time series.
- Every meter carries the common `region` tag.
- `@Observed` produced the timer `order_lookup_seconds` without any `MeterRegistry` code.

Push vs pull. Prometheus **pulls** (scrapes) `/actuator/prometheus` every 15–60 s; other backends are **pushed** to, e.g. OTLP via `management.otlp.metrics.export.url=http://collector:4318/v1/metrics`. In `@SpringBootTest` Boot switches metric exporters off (only an in-memory registry) unless the class has `@AutoConfigureMetrics` — hence its use in the Prometheus test.

Cardinality — the one rule that matters

Cardinality = the number of distinct tag-value combinations, each a separate stored time series. Tags need a **small, bounded** value set (method, status, region, result). Never tag with user ids, order ids, emails, full URLs or timestamps — a million users means a million series, which can crash Prometheus or explode a SaaS monitoring bill. Put such values in **traces or logs**.

Common mistakes

- High-cardinality tags (above).
- `Timer.builder(...).register(...)` in a hot loop is fine (returns the cached meter), but *dynamic meter names* are not — use tags.
- Averages for latency — use percentiles (p95, p99: the time 95%/99% of requests stay under; System Design Fundamentals) from histograms; averages hide the slow tail users feel.
- Gauges on garbage-collectable objects: `registry.gauge(name, obj, fn)` holds a **weak** reference (doesn't stop the object being freed); if nothing else holds `obj`, the gauge reports `NaN`.

15.6 The Observation API and `@Observed`

What it is

An **Observation** (Micrometer 1.10+, Boot 3+) wraps a unit of work — "look up an order" — once, and **handlers** decide what to produce: a `Timer` (metrics handler), a span (tracing handler), a log line, anything. Spring's own instrumentation (MVC, WebFlux, `RestClient`, `WebClient`, JDBC via extensions, Spring Security, Kafka, `@Scheduled`...) is written this way, so one dependency change yields both metrics and traces.

```
Observation.createNotStarted("order.lookup", observationRegistry)
        .lowCardinalityKeyValue("channel", "web")     // -> metric tag AND span attribute
        .highCardinalityKeyValue("orderId", "42")     // -> span attribute only
        .observe(() -> service.find(42));
```

- **Low-cardinality key values** become metric tags *and* span attributes.
- **High-cardinality key values** go only to spans (safe for ids).

`@Observed(name = "order.lookup", contextualName = "order-lookup")` on a bean method does the same via an aspect (Chapter 4). It needs the AspectJ starter (`spring-boot-starter-aspectj` in Boot 4) and, since Boot 3.2, the property:

```
management.observations.annotations.enabled=true   # also activates @Timed, @Counted, @NewSpan
```

The metric is named after `name` (`order.lookup` → `order_lookup_seconds_*` above, tagged `class` and `method`); the span is named after `contextualName`:

```
spans for GET /orders/2 -> [http get /orders/{id}, order-lookup]
```

Tuning which observations happen

```
# switch off whole families by name prefix
management.observations.enable.spring.security=false
# common key values added to every observation
management.observations.key-values.region=eu-west-1
```

Spring Security creates several observations per request (filter chain before/after, authentication, authorization); in the demo's first run they were most of the exported spans, so disabling the `spring.security` prefix keeps traces readable. For finer control register an `ObservationPredicate` bean.

Context propagation. Observations live in thread-local state: for `@Async` on Boot's executor set `spring.task.execution.propagate-context=true` (or a `ContextPropagatingTaskDecorator` on your own executor); for Reactor `spring.reactor.context-propagation=auto`.

15.7 Distributed tracing

What it is

A **trace** follows one request across every service it touches: a tree of **spans**, each one timed operation (incoming request, outgoing call, DB query) with a **span id**, the tree's shared **trace id**, a **parent span id** (`0000000000000000` for the root), start/end time and attributes — like a parcel tracking number every depot scans, adding its own entry.

How it works

1. A request arrives without trace headers → the server observation starts a **root span** with a new 128-bit trace id.
2. The trace context is stored in the current thread and copied into the log MDC (`traceId`, `spanId`).
3. An outgoing call through an instrumented client (Boot's `RestClient.Builder`, `WebClient.Builder`, `RestTemplateBuilder`, Kafka templates...) creates a **client span** and injects the context into headers — by default the W3C **Trace Context** header `traceparent: 00-<traceId>-<spanId>-01`.
4. The downstream service reads the header and continues the same trace with a child span.
5. Finished spans go to a **span processor** → **exporter** (OTLP to a collector, Zipkin, ...). Boot batches them (`BatchSpanProcessor`).
6. The backend (Tempo, Jaeger, Zipkin) stitches spans by trace id into a timeline.

Sampling decides which traces are recorded: `management.tracing.sampling.probability` (default **0.1** = 10%). The decision is made at the root and travels in the header, so a trace is either complete or absent.

Setup in Boot 4

```
<!-- Micrometer Tracing + OpenTelemetry bridge + OTLP exporters -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-opentelemetry</artifactId>
</dependency>
<!-- tests: spring-boot-starter-opentelemetry-test (provides @AutoConfigureTracing) -->
```

```
management.tracing.sampling.probability=1.0     # 100% in dev/test; lower in high-traffic prod
# send spans to an OpenTelemetry Collector (not used in the demo - no collector running)
# management.opentelemetry.tracing.export.otlp.endpoint=http://otel-collector:4318/v1/traces
```

The Brave/Zipkin alternative is `spring-boot-starter-zipkin` with `management.tracing.export.zipkin.endpoint`. Boot 3 users: the dependency used to be `micrometer-tracing-bridge-otel` + `opentelemetry-exporter-otlp`; the one starter replaces both.

A "logging exporter" instead of a collector

To see spans without running Jaeger, the demo registers its own `SpanExporter` bean; Boot adds every `SpanExporter` bean to its batch processor:

```
@Component
public class PrintingSpanExporter implements SpanExporter {
    private final List<SpanData> finished = new CopyOnWriteArrayList<>();

    @Override
    public CompletableResultCode export(Collection<SpanData> spans) {
        finished.addAll(spans);                      // a real exporter would send them over OTLP
        return CompletableResultCode.ofSuccess();
    }
    // ... flush(), shutdown(), finished(), clear()
}
```

The controller calls a second endpoint over HTTP with Boot's `RestClient.Builder`, standing in for a downstream service:

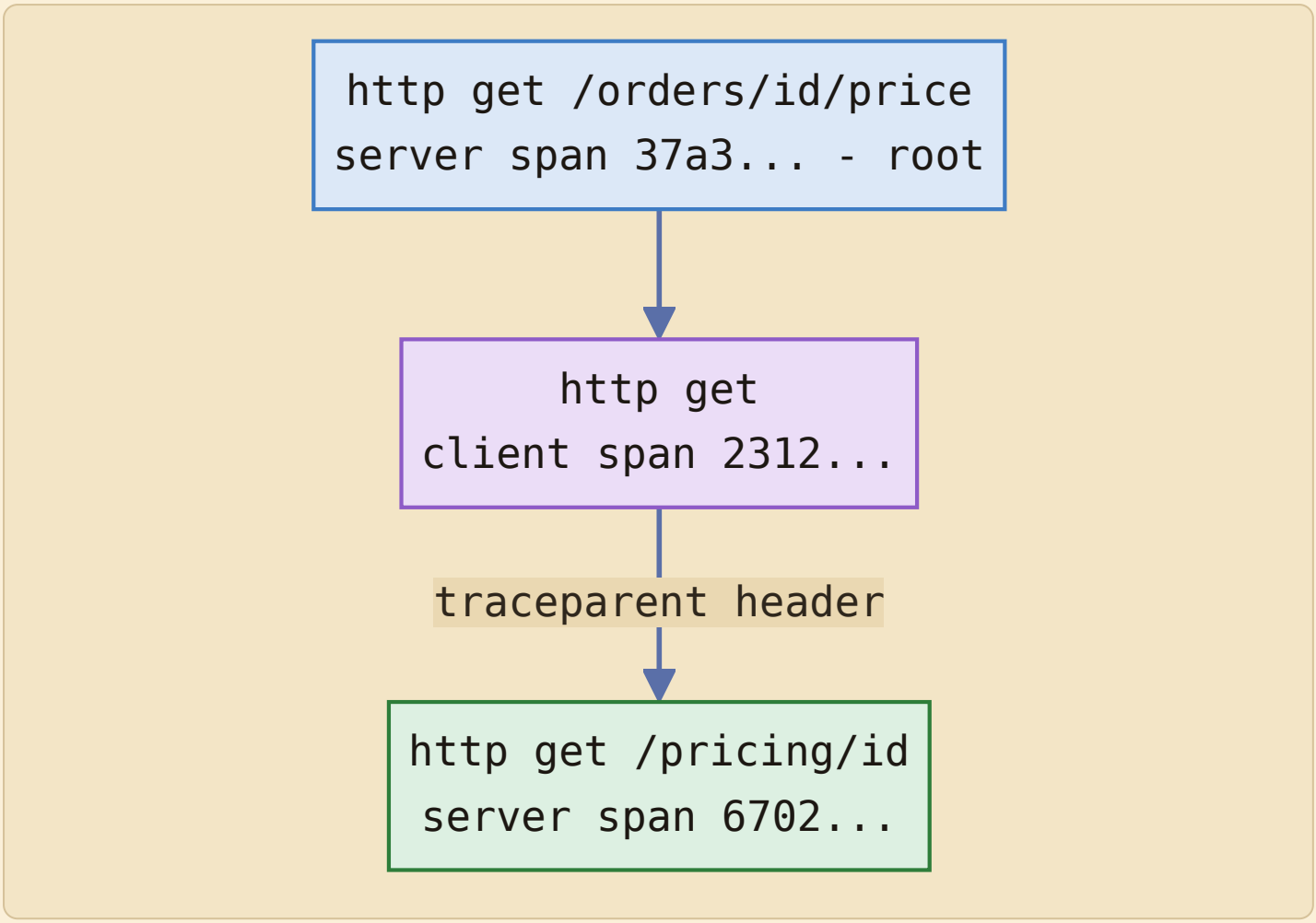
```
@GetMapping("/orders/{id}/price")
public String price(@PathVariable long id) {
    log.info("asking pricing service for order {}", id);
    String base = "http://localhost:" + env.getProperty("local.server.port");
    return restClientBuilder.build().get()
            .uri(base + "/pricing/{id}", id)
            .retrieve().body(String.class);
}

@GetMapping("/pricing/{id}")
public String pricing(@PathVariable long id) {
    log.info("pricing order {}", id);
    return "price for " + id + " = 49.99";
}
```

`TracingTests` (with `@AutoConfigureTracing`, because tests disable span export by default) calls `/orders/1/price`, flushes the batch processor and prints the log lines and spans:

```
response: price for 1 = 49.99
log lines (trace id - span id in brackets):
  [orders] [o-auto-1-exec-8] [89fb1d6934d0cf54e0910695a8df7859-37a3311dfdcc61c3] ch15.obs.OrderController
: asking pricing service for order 1
  [orders] [o-auto-1-exec-9] [89fb1d6934d0cf54e0910695a8df7859-6702c6550d5a9bfd] ch15.obs.OrderController
: pricing order 1
finished spans:
  http get /orders/{id}/price  trace=89fb1d6934d0cf54e0910695a8df7859 span=37a3311dfdcc61c3
parent=0000000000000000
  http get                     trace=89fb1d6934d0cf54e0910695a8df7859 span=2312285359b3b845
parent=37a3311dfdcc61c3
  http get /pricing/{id}       trace=89fb1d6934d0cf54e0910695a8df7859 span=6702c6550d5a9bfd
parent=2312285359b3b845
```

Read it as a tree:



- Two Tomcat threads (`exec-8`, `exec-9`) logged with the **same trace id** — propagation via `traceparent`.
- Each log line's span id matches the active span (`37a3...` outer request, `6702...` downstream).
- Boot's default console pattern prints `[application-name] [thread] [traceId-spanId]`; change it with `logging.pattern.correlation`.
- Across *real* services the picture is identical; only the exporter sends spans to a shared backend.

Baggage is extra key/value context travelling with the trace (e.g. a tenant id): `management.tracing.baggage.remote-fields=tenant` sends it in headers, `management.tracing.baggage.correlation.fields=tenant` also puts it in the MDC.

Trade-offs

Choice	Pro	Con
Sampling 100%	every request traceable	CPU, network and storage cost at scale
Head sampling (e.g. 10%, Boot default)	cheap, simple	the rare failing request is often not sampled
Tail sampling in the OTel Collector	keep all errors/slow traces, drop the rest	collector must buffer whole traces
OTel **Java agent** (a jar attached at JVM start that auto-instruments classes) instead of Micrometer	zero code, instruments many libraries	a second instrumentation layer; version coupling with the agent

Common mistakes

- `RestClient.builder()` / `new RestTemplate()` built yourself → no instrumentation; the trace breaks at that hop. Inject Boot's builder.
- Context lost on your own thread pools or `CompletableFuture` → wrap executors with context propagation.
- Trace ids as metric tags → cardinality explosion.
- Sampling 1.0 in busy production without a storage plan.

In the interview

Q: How does distributed tracing work in Spring Boot? 30-second answer: Micrometer Observation instruments incoming and outgoing calls; Micrometer Tracing with the OpenTelemetry (or Brave) bridge turns observations into spans. A trace id is created at the edge, propagated in the W3C `traceparent` header by instrumented clients, and put in the log MDC. Spans are batched and exported via OTLP to a collector and viewed in Tempo/Jaeger/Zipkin. Sampling defaults to 10%.

- *Trace broken between two services?* → Client not instrumented (created with `new`), a proxy strips headers, propagation formats differ (W3C vs B3, Zipkin's older format), or async code lost the context.
- *Logs vs traces?* → Traces show timing and causality across services; logs give detail within one; the shared trace id links them.

TL;DR

`spring-boot-starter-opentelemetry` + `management.tracing.sampling.probability`; inject Boot's client builders; trace/span ids appear in logs automatically; export via `management.opentelemetry.tracing.export.otlp.endpoint`.

15.8 Structured logging

What it is

Structured logging writes each event as a machine-readable record (JSON) instead of a formatted line, so log platforms (Elasticsearch/Kibana, Loki, Graylog, Datadog, Cloud Logging) filter on fields — `log.level:ERROR AND service.name:orders AND traceId:...` — instead of regex-parsing text. Built into Boot since **3.4**.

```
logging.structured.format.console=ecs        # ecs | logstash | gelf
logging.structured.format.file=logstash      # file output can use a different format
logging.structured.ecs.service.version=1.0.0
logging.structured.ecs.service.environment=test
```

Format	Standard	Typical destination
`ecs`	Elastic Common Schema	Elasticsearch / Elastic Agent, also fine for Loki
`logstash`	Logstash JSON	Logstash, Fluentd/Fluent Bit
`gelf`	Graylog Extended Log Format	Graylog

`StructuredLoggingTests` starts the app with `ecs` and captures the log line of one request:

```
ECS log line:
{"@timestamp":"2026-09-28T23:50:30.319867Z","log":{"level":"INFO","logger":"ch15.obs.OrderService"},"process":
{"pid":17381,"thread":{"name":"http-nio-auto-2-exec-1"}},"service":
{"name":"orders","version":"1.0.0","environment":"test","node":{}},"message":"looking up order
1","traceId":"636cb6141b8e904e29a6f7824c3eb713","spanId":"8ceaaf80e43135b1","ecs":{"version":"8.11"}}
```

`service.name` comes from `spring.application.name`; MDC entries (here `traceId`, `spanId`) become fields automatically, as do key/values from SLF4J's fluent API (`log.atInfo().addKeyValue("orderId", id).log("placed")`).

Customise the JSON without code:

```
logging.structured.json.exclude=process.pid
logging.structured.json.rename.process.thread.name=thread
logging.structured.json.add.team=checkout
logging.structured.json.stacktrace.root=first          # root cause first
logging.structured.json.stacktrace.max-throwable-depth=20
```

For full control implement `StructuredLogFormatter<ILoggingEvent>` and name the class in `logging.structured.format.console`.

When not to use it: local development in a terminal (JSON is hard to read). Common setup: text by default, JSON in the `prod` profile (`application-prod.properties`), or JSON to file and text to console.

Common mistakes: logging secrets or personal data (JSON makes them *easier* to search — mask them); multi-line text stack traces that the shipper (the agent sending logs to the platform) splits into many events (JSON keeps one field); DEBUG on hot paths in production. Practical catch: logging is global to the JVM, so the structured-logging test runs last (`junit-platform.properties` orders the classes).

15.9 Graceful shutdown

What it is

When an instance stops (deployment, scale-down, node drain), **in-flight** requests should finish, not fail. **Graceful shutdown** stops accepting new connections, waits for active requests up to a timeout, then stops. **Default since Boot 3.4** for Tomcat, Jetty and Reactor Netty (`server.shutdown=graceful`); `server.shutdown=immediate` turns it off.

```
spring.lifecycle.timeout-per-shutdown-phase=10s   # default 30s
```

How it works

1. The JVM receives `SIGTERM` — the OS's polite "please stop", sent by Kubernetes and `docker stop` (Ctrl+C sends the similar `SIGINT`); unlike `SIGKILL` it runs shutdown hooks → Spring's hook closes the context.
2. Readiness switches to `REFUSING_TRAFFIC`.
3. The web server's lifecycle bean stops first: it stops accepting new connections at the network level and waits for active requests.

4. When they finish (or the timeout expires), the rest of the context closes: `@PreDestroy`, connection pools, executors.

Proof

`GracefulShutdownTests` starts the app, begins a `/slow?ms=1000` request, closes the context mid-request (as `SIGTERM` would), and tries a new request 200 ms later — once per mode:

```
ConfigurableApplicationContext ctx = SpringApplication.run(RuntimeApplication.class,
        "--server.port=0", "--server.shutdown=" + shutdownMode, "--spring.main.banner-mode=off");
// ... start a 1-second request, wait until it is inside the controller ...
Thread closer = Thread.ofPlatform().start(ctx::close);        // like SIGTERM from Kubernetes
Thread.sleep(200);
// ... try a new request, then wait for both ...
```

```
server.shutdown=immediate in-flight -> FAILED: IOException | new request during shutdown -> rejected
(ConnectException) | close took < 0.7 s (did not wait)
server.shutdown=graceful  in-flight -> HTTP 200 'done after 1000 ms' | new request during shutdown -> rejected
(ConnectException) | close took ~1 s (waited)
  log: Commencing graceful shutdown. Waiting for active requests to complete
  log: Graceful shutdown complete
```

In Kubernetes

A race: Kubernetes sends `SIGTERM` and removes the pod from the Service endpoints **at the same time**, and load balancers may route new requests for a few seconds. Since the server refuses new connections immediately (connection refused, not 503), delay shutdown briefly so endpoint removal propagates:

```
spec:
  terminationGracePeriodSeconds: 45          # must exceed preStop + Spring timeout
  containers:
    - name: orders
      lifecycle:
        preStop:
          sleep: { seconds: 10 }             # K8s 1.30+; older: exec ["sh","-c","sleep 10"]
```

Common mistakes

- `timeout-per-shutdown-phase` longer than `terminationGracePeriodSeconds` (default 30 s) → the pod is `SIGKILL`ed mid-drain.
- Expecting graceful shutdown to finish **message consumers** or `@Async` jobs — configure those separately (listener container stop, `spring.task.execution.shutdown.await-termination=true`).
- Running from an IDE that kills the process instead of sending SIGTERM (shutdown then looks immediate).

15.10 Virtual threads

Virtual threads (final in Java 21, Java book 15.4) are cheap JVM-managed threads; a blocked one releases its carrier OS thread, so blocking servlet apps hold many more concurrent requests without a huge pool. In Boot (3.2+):

```
spring.threads.virtual.enabled=true
```

Tomcat/Jetty then run each request on a virtual thread, and Boot's `@Async` executor, `@Scheduled` scheduler and several messaging listener containers switch too (14.5). `VirtualThreadsTests` calls an endpoint reporting `Thread.currentThread()`:

```
spring.threads.virtual.enabled=true  -> virtual thread: VirtualThread[#100,tomcat-handler-
0]/runnable@ForkJoinPool-1-worker-1
default                              -> platform thread: Thread[#108,http-nio-auto-4-exec-1,5,main]
```

Caveats.

- The **bottleneck moves**: 10,000 concurrent requests now hit the database pool (10 connections by default) or a downstream API. Limit concurrency explicitly — `@ConcurrencyLimit` (4.7), pool sizes, bulkheads.
- Pinning: before JDK 24 blocking inside `synchronized` pinned the carrier; JDK 24 (JEP 491) fixed that; native calls can still pin.
- Virtual threads are daemon threads: an app with only `@Scheduled` work may exit; set `spring.main.keep-alive=true`.
- CPU-bound work gains nothing. Virtual threads vs reactive: Chapter 13.

15.11 Faster startup: extracted jars, CDS, AOT cache, native images

The problem it solves

A typical Boot app takes 1–5 s to start on the JVM (class loading, reflection, bean creation, JIT warm-up). That matters for autoscaling, serverless (scale-to-zero: no instance until a request arrives) and fast rollbacks. Options, least to most invasive:

Technique	What it does	Startup gain	Cost
Extracted jar	run from unpacked `lib/` instead of nested jars	small	a build step
CDS (Class Data Sharing; application classes since JDK 10, dynamic archives since JDK 13)	archive of parsed classes from a training run (a short run that records what loads)	~20–40%	training run, same JDK/classpath
JDK AOT cache (JEP 483 in JDK 24, simpler flags JEP 514 and method profiles JEP 515 in JDK 25)	caches classes *loaded and linked* (plus profiles) from a training run	often ~2×	training run; cache tied to the JDK build and exact jars
Spring AOT on the JVM	generates bean-definition code at build time (`spring-boot:process-aot`), run with `-Dspring.aot.enabled=true`	modest	profiles/conditions fixed at build time
GraalVM native image	compiles the whole app ahead of time to a native executable	~50–100× (tens of ms)	long builds, closed-world limits, reflection hints, lower peak throughput without PGO (profile-guided optimisation)

Measured: AOT cache on JDK 25

`aot-cache-demo.sh` builds `production-runtime`, extracts it and times start-until-context-refreshed (`-Dspring.context.exit=onRefresh` exits right after refresh — a Framework 6.1+ feature for training runs):

```
java -Djarmode=tools -jar target/app.jar extract --destination target/app   # app jar + lib/
cd target/app
# training run: record and write the cache in one step (JDK 25)
java -XX:AOTCacheOutput=app.aot -Dspring.context.exit=onRefresh -jar app.jar
# production run
java -XX:AOTCache=app.aot -jar app.jar
```

Run as `bash aot-cache-demo.sh <JDK-25-home>` (else `$JAVA_HOME`). Rows 2 and 3 are medians of 5 runs (Apple silicon laptop, Homebrew OpenJDK 25.0.2; **machine-dependent** — measure your own):

```
java: openjdk version "25.0.2" 2026-01-20
1. fat jar (java -jar)               : 1152 ms (mean of 3)
2. extracted jar                     : 934 ms
   AOT cache file                    :  56M
3. extracted jar + -XX:AOTCache      : 467 ms
```

The AOT cache more than halved JVM-to-context-ready time here, for a 56 MB file and a training step. CDS on JDK 13–24: `-XX:ArchiveClassesAtExit=app.jsa` (training), `-XX:SharedArchiveFile=app.jsa`. The cache is valid only for the same JDK build and jars (it needs a plain classpath — hence extracting the nested-jar fat jar); otherwise the JVM ignores it with a warning.

GraalVM native image (not run — `native-image` is not installed here)

```
# needs GraalVM / Liberica NIK 25; spring-boot-starter-parent provides the "native" profile
mvn -Pnative native:compile          # -> target/<artifactId>, a native executable
mvn -Pnative spring-boot:build-image # same, inside a container image (needs Docker)
```

Plus `org.graalvm.buildtools:native-maven-plugin` in the POM. The build runs Spring AOT, then GraalVM analyses everything reachable from `main` (**closed world**): classes unseen at build time can't load later; reflection, resources, proxies and serialization need **hints** (Spring generates most; add yours with `RuntimeHintsRegistrar` or `@RegisterReflectionForBinding`). Profiles and `@ConditionalOnProperty` are evaluated at build time. Test the native build in CI (`mvn -PnativeTest test`) — some libraries fail only there.

Choose native for CLIs, serverless functions, scale-to-zero and memory-tight deployments. **Stay on the JVM** (with the AOT cache, which keeps the JIT) for long-running, throughput-heavy services where peak performance and easy debugging win.

Container images (not run — Docker is not running here)

Cloud Native **Buildpacks** turn the jar into an **OCI image** (the standard container-image format) with no Dockerfile:

```
mvn spring-boot:build-image -Dspring-boot.build-image.imageName=acme/orders:1.0
# JVM options for the buildpack, e.g. enable the AOT cache or CDS training at build time:
#   <env><BP_JVM_AOTCACHE_ENABLED>true</BP_JVM_AOTCACHE_ENABLED></env>   (or BP_JVM_CDS_ENABLED)
```

Or a layered Dockerfile (each instruction a cached layer) — dependencies change rarely, so they get their own cached layer:

```
FROM eclipse-temurin:25-jre AS build
WORKDIR /build
COPY target/orders.jar app.jar
RUN java -Djarmode=tools -jar app.jar extract --layers --destination out

FROM eclipse-temurin:25-jre
WORKDIR /app
COPY --from=build /build/out/dependencies/ ./
COPY --from=build /build/out/spring-boot-loader/ ./
COPY --from=build /build/out/snapshot-dependencies/ ./
COPY --from=build /build/out/application/ ./
RUN java -XX:AOTCacheOutput=app.aot -Dspring.context.exit=onRefresh -jar app.jar
USER 1000
ENTRYPOINT ["java", "-XX:AOTCache=app.aot", "-XX:MaxRAMPercentage=75", "-jar", "app.jar"]
```

`-XX:MaxRAMPercentage` sizes the heap from the container's memory limit; running as a non-root user is basic hardening.

15.12 Production configuration checklist

Area	Setting / practice
Actuator exposure	expose only what you use (`health,info,prometheus`); protect the rest with `EndpointRequest`; consider `management.server.port`
Health	probes (on by default in Boot 4) wired to Kubernetes; custom indicators fast; no dependencies in liveness
Metrics	Prometheus or OTLP registry; common tags (`application`, `region`); histograms for latency SLOs (targets like "p99 under 300 ms"); bounded tag values
Tracing	`spring-boot-starter-opentelemetry`; sampling rate chosen consciously; OTLP endpoint to a collector
Logging	structured JSON in prod profile; levels INFO/WARN; no secrets or personal data; change levels at runtime via `/actuator/loggers`
Shutdown	graceful (default) + `timeout-per-shutdown-phase` < K8s grace period; `preStop` delay
Threads	`spring.threads.virtual.enabled=true` for blocking I/O apps, with explicit concurrency limits
Startup	extracted jar + AOT cache (JDK 25) or native image where startup matters
Config & secrets	externalised config per environment (Chapter 3); secrets from a vault or K8s secrets, never in the jar
Resources	container memory limit + `MaxRAMPercentage`; connection pool sized to the database
Errors	`spring.web.error.include-stacktrace=never` (default; Boot 4 name — the old `server.error.*` keys are no longer bound since 4.0.0) and Problem Details (Chapter 5)

15.13 Common mistakes (whole chapter)

Symptom	Cause	Fix
`/actuator/prometheus` returns 404	registry dependency missing or endpoint not exposed	add `micrometer-registry-prometheus`, include `prometheus` in exposure
Prometheus endpoint missing only in tests	Boot disables metric export in tests	`@AutoConfigureMetrics`
No trace ids in logs	no tracer on the classpath, or a custom log pattern without `%X{traceId}`	add `spring-boot-starter-opentelemetry`; keep Boot's correlation pattern
Trace breaks between services	client created with `new` / `builder()`	inject Boot's `RestClient.Builder` / `WebClient.Builder`
`@Observed` does nothing	property or AspectJ starter missing, or self-invocation	`management.observations.annotations.enabled=true`, `spring-boot-starter-aspectj`, call through the proxy
Prometheus memory explodes	high-cardinality tags	remove ids from tags
Pods restart during DB outage	DB in liveness	keep liveness internal
502s (Bad Gateway) during deployments	no `preStop` delay, grace timeout too short	`preStop` sleep, align timeouts
Secrets visible in `/actuator/env`	`show-values=always`	leave the default `never`, secure the endpoint

---

15.14 In the interview

Q: What is Spring Boot Actuator and how do you secure it? Management endpoints (health, info, metrics, prometheus, loggers, env, ...). Only `health` is exposed by default; expose more with `management.endpoints.web.exposure.include`, restrict with `management.endpoint.<id>.access`, and secure with a `SecurityFilterChain` using `EndpointRequest.to(...)` / `toAnyEndpoint()`, often on a separate management port.

- *Change a log level in production without redeploying?* → `POST /actuator/loggers/<logger>` with `{"configuredLevel":"DEBUG"}` (protected endpoint).

Q: Counter, Gauge, Timer — when to use which? → Counter for accumulating events (rate via PromQL `rate()`), Gauge for current values (queue depth), Timer for durations (count/sum/max/histogram → latency percentiles). Keep tags low-cardinality.

Q: What does the Observation API add over plain Micrometer metrics? → Instrument once, get metrics *and* traces (and custom handlers); Spring's own instrumentation uses it; `@Observed` for methods.

Q: How does a trace id get from service A to B? → The instrumented HTTP client injects `traceparent`; B's server observation extracts it and creates a child span; both log the same trace id via MDC.

Q: Zero-downtime deployments with Spring Boot on Kubernetes? → Readiness probe (ready after startup), graceful shutdown (default), `preStop` delay, grace period longer than the drain timeout, rolling update with `maxUnavailable: 0`. See blue-green and canary.

Q: How would you speed up startup? → Lazy init (beans created on first use) where safe, fewer auto-configurations, extracted jar + CDS/AOT cache (JDK 25: `-XX:AOTCacheOutput` training, `-XX:AOTCache`), Spring AOT, or GraalVM native image — trading build complexity and closed-world limits for startup.

Q: Virtual threads in Spring Boot? → `spring.threads.virtual.enabled=true` (Java 21+, Boot 3.2+); great for blocking I/O; move concurrency limits to explicit places (pools, `@ConcurrencyLimit`).

Chapter TL;DR

- Observability = **logs + metrics + traces**, linked by the trace id. Spring: Logback, **Micrometer**, **Micrometer Tracing + OpenTelemetry**, all fed by the **Observation API**.
- **Actuator**: only `health` exposed by default; control with `exposure.include` and `access`; secure with `EndpointRequest` (Boot 4 package `org.springframework.boot.security.autoconfigure.actuate.web.servlet`).
- **Health**: `HealthIndicator` (Boot 4 package `org.springframework.boot.health.contributor`), groups, **liveness/readiness on by default in Boot 4**; DOWN → 503; no external deps in liveness.
- **Metrics**: Counter/Timer/Gauge with **low-cardinality** tags; `/actuator/prometheus` for scraping; `@AutoConfigureMetrics` in tests.
- **Observation / `@Observed`**: one instrumentation → timer + span; needs `management.observations.annotations.enabled=true` + AspectJ.
- **Tracing**: `spring-boot-starter-opentelemetry`; W3C `traceparent` propagation via Boot's client builders; `[traceId-spanId]` in logs; sampling default 0.1; export over OTLP.
- **Structured logging** (Boot 3.4+): `logging.structured.format.console=ecs|logstash|gelf`; MDC fields included.
- **Graceful shutdown** is the default (Boot 3.4+); tune `spring.lifecycle.timeout-per-shutdown-phase`; add a `preStop` delay on Kubernetes.
- **Virtual threads**: `spring.threads.virtual.enabled=true`.
- **Startup**: extract the jar, train an **AOT cache** (JDK 25; 934 → 467 ms here), or build a **GraalVM native image**; package with buildpacks or a layered Dockerfile.
- Next: Chapter 16 — Spring Cloud and Microservices.